What this tool does

GLOF Risk Tool — a research prototype that screens 14 Himalayan glacial lakes for outburst-flood danger, and drafts the situation report that follows one. Plain language throughout. Every number is read from the tool's own output, not typed here.

The problem it is built for

High in the Himalaya, meltwater collects behind natural dams of loose rock and ice. Sometimes a dam gives way and the lake empties in hours. The flood can travel tens of kilometres, killing people and destroying bridges and hydropower plants far downstream.

The usual way to find the risky ones is to watch which lakes are **growing**, and to ignore any lake smaller than 0.1 km². In August 2024 a lake above the village of Thame, Nepal, burst. It was 0.020 km² — a fifth of that size threshold — and it had not grown. Size-based screening never looked at it.

Part 1 — the watcher: which lakes deserve attention

It reads free satellite pictures and elevation maps for 14 lakes, measures each lake in every usable image from 2017 to 2025, and then asks the questions that size misses:

- Is there steep, unstable ground above this lake?
- Could an avalanche from up there actually **reach** the water, or would it stop on the way?
- How much rock and ice could fall in, compared with how much water is already held back? **This is the main ranking measure.**
- How much freeboard has the dam got, how steep is its outer face — and where would the water go if it failed?

Nine of these checks come from published research, with the paper and a confidence level recorded beside every threshold. On pre-event data alone, Thame comes out **1 of 13** on the main measure.

Part 2 — the reporter: what to tell people afterwards

It reads news articles and official documents about flood disasters and drafts a situation report in English and Nepali, in the standard humanitarian format, plus machine-readable alert files. Its most useful habit is refusing to answer. Reports of the July 2025 Rasuwa flood put the number of destroyed hydropower plants at 4, 5, 8 and 11 — so it prints the whole range with a name against each figure, instead of picking one and sounding certain.

Every sentence is checked against the sources it cites and anything unsupported is struck before release; a named person must then type an approval, and a blank keypress is refused.

Part 3 — the map you can actually use

A single web page, opened from disk with no internet and no map server. Each lake's real measured outline sits on shaded terrain, with a slider to move through the years and watch it change, the modelled flood corridor, and a movable alarm threshold that re-screens all 14 lakes as you drag it. That last control is deliberate: the published alarm level was chosen after seeing every score, so the quickest way to show what depends on it is to let people move it. The ranking does not move. The flag count does.

Does it work?

Tested on four real Himalayan disasters, using only information that existed *before* each event:

	Size-only screening	This tool
Found the Thame lake?	No — never assessed it	Yes — ranked it 1 of 13

	Size-only screening	This tool
Burst lakes caught	0 of 3	1 of 3
Lake sizes measured accurately	—	6 of 8

And for the written reports, against an ordinary one-shot AI summary of the same sources:

	Ordinary AI summary	This tool
Made-up figures	34%	9%
Disagreements between sources spotted	0%	95%
Figures traceable to a named source	66%	91%

Knowing what is *not* a lake burst

In February 2021 a disaster at Chamoli in India killed over 200 people and was reported worldwide as a glacial lake burst. It was not — it was a rock-and-ice avalanche, with no lake involved. The tool finds 300 m^2 of scattered meltwater, below its own minimum, assesses nothing, and refuses to call it a lake burst in both languages.

This is not a historical curiosity. On 26 August 2026 a mass of rock and glacier ice fell from Langtang-Lirung in Rasuwa, dammed a river, and the dam burst — killing hundreds. The first question asked everywhere was whether a glacial lake had burst. It took days to establish that one had not. Getting that question right decides whether you go and watch lakes or watch slopes.

What it cannot do

These limits are part of the design. A screening tool whose weaknesses are hidden is more dangerous than no tool.

- **It cannot predict floods.** It does not model the trigger — the avalanche that decides the date is not observed. It ranks standing danger, so you know where to send someone with instruments.
- **It only screens lakes.** The Langtang disaster above began on a slope with no lake on it. This tool would have had nothing to say.
- It reads shape and surroundings from pictures — it cannot see inside a dam, so it cannot say how strong that dam is.
- Clouds hide lakes during the monsoon, which is when most floods happen: only 1 of 16 satellite passes in the days before these four disasters was clear enough to use.
- It measures 6 of 8 test lakes accurately — and two of those came from correcting our own reference data, not the measurement.
- Flood paths are rough corridors, not flood maps.
- **It is a research prototype.** It is not a warning system and must not be used to alert the public.

Who it is for

Nepal's own agencies — DHM, NDRRMA and ICIMOD — hold the data and the legal responsibility; this feeds their work rather than replacing it. Nepal already learned what over-automation costs: the Tsho Rolpa early-warning system of 2000–02, sirens across 19 villages, is now defunct. Anyone can re-run this analysis and get the same numbers — it ships with its data, needs no internet and no accounts, and produces byte-for-byte identical results on Windows and Linux alike.